

Dietary Supplementation of Octacosanol Improves Exercise-Induced Fatigue and Its Molecular Mechanism

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ABSTRACT: Several publications report that octacosanol (OCT) has different biological functions. This study was designed to evaluate the antifatigue effect and molecular mechanism of octacosanol (200 mg/(kg day)) in forced exercise-induced fatigue models of trained male C57BL/6 mice. Results showed that octacosanol ameliorated the mice's autonomic activities, forelimb grip strength, and swimming endurance, and the levels of liver glycogen (LG), muscle glycogen (MG), blood lactic acid (BLA), lactate dehydrogenase (LDH), superoxide dismutase (SOD), and glutathione peroxidase (GSH-Px) were also regulated. Gene analysis results showed that treatment with OCT upregulated 29 genes, while 38 genes were downregulated in gastrocnemius tissue. Gene ontology (GO) analyses indicated that these genes enriched functions in relation to myofibril, contractile fiber, and calcium-dependent adenosinetriphosphatase (ATPase) activity. Octacosanol supplementation significantly adjusted the messenger RNA (mRNA) and protein expression levels related to fatigue performance. Octacosanol has an observably mitigating effect in exercise-induced fatigue models, and its molecular mechanism may be related to the regulation of tripartite motif-containing 63 (Trim63), periaxin (Prx), calcium voltage-gated channel subunit $\alpha 1$ H (Cacna1h), and myosin-binding protein C (Mybpc3) expression.

KEYWORDS: *antifatigue, octacosanol, energy and endurance performance, biochemical parameters, microarray, molecular mechanism*

INTRODUCTION

Fatigue is one of the most common physiological reactions related to a progressive gradual decline in the ability for vitality and activity.¹ Fatigue can either be physiologic or chronic. Physiological fatigue is the disability of muscles to maintain normal movements due to severe stress and intensive physical or mental activity.² Therefore, exercise fatigue is a distinguishingly physiological process after strenuous exercise that involves a significant decrease in the ability for more work, vitality, and activities.³ This situation mainly results from the depletion of energy sources and the accumulation of end products that eventually lead to the disorder in the internal environment of the body.⁴ Besides from physiological reaction to physical stress, fatigue is also indicative of disorders, such as immunologic, hematological, endocrinological diseases, etc.⁵ Fatigue also progresses in aging, cancer, depression, multiple sclerosis, Parkinson's, etc.⁶ Accelerated life pace and fierce competitive social lifestyles are making fatigue a commonly occurring condition. Reports abound of some traditional/natural medicines and food constituents used as healthcare products for the relieve of body fatigue include *Antrodia camphorata*,² *Radix Rehmanniae Preparata* (Shu Dihuang),⁴ Loach protein,⁵ dietary nucleotides,⁷ etc. Additionally, medical practitioners often advise one to get enough sleep and avoid staying up late. With these arrays of options for fatigue alleviation, it is still necessary for patients affected by the above-mentioned associated diseases to take medication-based therapies.⁶ Synthetic drugs or therapies currently available for fatigue treatment have not been top-notch, and hence, efforts

are directed toward nutritional interventions that are safer and effective fatigue prevention methods.⁷ Therefore, it is imperative to investigate natural antifatigue compounds,⁸ most of which could improve athletic ability, accelerate the elimination of fatigue by-products in humans, without attendant adverse effects.

Natural products have no side effects or cause very few side effects. They have attracted the attention of many researchers in the field of biomedicine. Many active ingredients have been used in the development of drugs and functional foods. Octacosanol (OCT) is a regular monohydric high-grade saturated linear aliphatic alcohol and an important component of a natural wax extracted from multiple plants and animals.^{9,10} It has a diet of interest due to its excellent biological activities; of interest was the identification of OCT as a functional antifatigue agent, which could enhance endurance and energy,¹¹ dietary and pharmaceutical supplements,¹² and as feed additives for animals.¹³ In addition, OCT has an anti-Parkinsonian effect,¹⁴ regulating cardiovascular functions,¹⁵ anti-inflammatory and antinociceptive effects,¹⁶ etc. As a natural compound, only a small portion gets ingested from diets. Adding OCT to diet will benefit people's health.⁹ In

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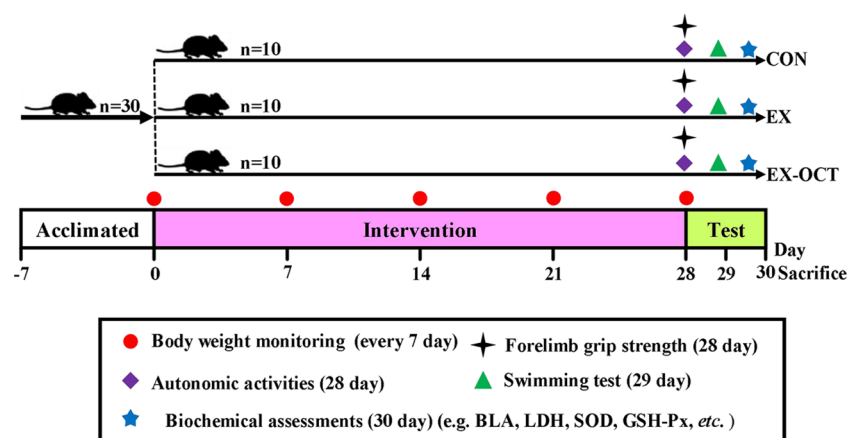


Figure 1. Experimental scheme for the effects of octacosanol on exercise adaptation. C57BL/6 mice were randomly assigned to the three groups ($n = 10$ per group) indicated (CON, EX, and EX-OCT) and were consecutively supplemented with octacosanol (200 mg/(kg day)) until the end of the experiments. The physical capabilities and related biochemistry properties of mice were evaluated during the test duration.

recent years, OCT also has other novel physiological functions, including improvement of the stress response, sleep quality, regulated lipid metabolism, and constipation relief.^{17–19} Nevertheless, most studies about OCT only evaluated its physiological functions and the molecular mechanism is poorly understood.

In our previous work, we found that OCT attenuated dextran sulfate sodium (DSS)-induced colitis in the mice and explained the molecular mechanisms of OCT in the mouse colitis model.²⁰ In this study, the mouse over-exercise model was used to evaluate the antifatigue effect of OCT. Gene chip analysis, bioinformatics, western blots, and real-time quantitative polymerase chain reaction (RT-qPCR) techniques were used in the experiments. We are the first to obtain the gene expression profile affected by OCT and identify the target genes and signal pathways regulated by OCT in the muscle tissues of the mouse over-exercise model. Bioinformatics indicated the protein–protein interaction network in the muscle cells after the addition of OCT to the diet. Our study preliminarily elucidates the antifatigue effect of OCT.

MATERIALS AND METHODS

Chemicals and Reagents. OCT (purity 99%, flakelike white powder) was brought from Sigma-Aldrich Company (St. Louis, MO). Liver glycogen (LG), muscle glycogen (MG), blood lactic acid (BLA), lactate dehydrogenase (LDH), superoxide dismutase (SOD), and glutathione peroxidase (GSH-Px) assay kits were purchased from Jiangsu Meimian Industrial Co., Ltd. (Jiangsu, China). Radio-immunoprecipitation assay (RIPA) lysis buffer was purchased from Servicebio (Servicebio Biotechnology Co. Ltd., Wuhan, China, Cat. G2002). Polyclonal antibody periaxin (Prx) (NBP2-93522), tripartite motif-containing 63 (Trim63) (AF5366-SP), calcium channel, voltage-dependent, T type, α 1H subunit (Cacna1h) (NBP1-22444SS), myosin-binding protein C (Mybpc3) (AB133499), and glyceraldehyde-3-phosphate dehydrogenase (GAPDH) (no. AT000) were brought from Cell Signaling Technology Co. Ltd. (Beverly, MA). Antimouse immunoglobulin G (IgG) horseradish peroxidase (HRP) conjugate (W4021), and anti-rabbit IgG HRP conjugate (V7951) were obtained from Promega Company (Madison, WI). All other chemical reagents of this investigation were of analytical grade.

Experimental Animals and Diet. For the care and treatment of experimental mice, we followed the guidelines for the ethical treatment of laboratory animals. We used 7-week-old special pathogen-free male C57BL/6 mice brought from Hunan SJA Laboratory Animal (SLAC, Co., Ltd., Changsha, Hunan, China). All mice were fed with a standard diet in the animal laboratory with

12/12 h reverse light/dark cycles at a stationary temperature of 23 ± 2 °C, relative humidity of $60 \pm 4\%$, and noise less than 48 dB throughout the experimental period. Before the experiment, mice were fed with a standard laboratory diet (32% carbohydrates, 22% proteins, 10% lipids, and 3% vitamins) (SLAC, Co., Ltd., Changsha, Hunan, China) and tap water. Ethical regulations of Hunan province on animal experiments and guidelines for the care and use of laboratory animals were followed in the study. The use of animals and experimental agreements were approved by the Guidelines for the Care and Use of Experimental Animals, the Hunan Prima Pharmaceutical Research Center Co., Ltd. (69181714-X, Liuyang, Changsha, Hunan, China). The investigation was approved by the office of Animal Experiment Ethics at the Central South University of Forestry and Technology.

Mouse Grouping. The animals were adapted to the new environment for 1 week before used in the study (Figure 1). During the first week, animals were trained to accustom themselves to swim twice (5 min per time), and the mice that failed at swimming were screened out in the period. After that, 30 mice were randomly divided into three groups ($n = 10$ per group): control (CON) group (200 mg distilled water/kg body of weight with no exercise), exercise (EX) group (200 mg distilled water/kg body of weight with exercise), and exercise with OCT (EX-OCT) group (200 mg distilled water suspension of OCT/kg body of weight with exercise).¹⁷ We prepared OCT gavage solution by mixing 50 mg of OCT (purity 99%) to 5 mL distilled water (contained 1% carboxymethyl cellulose sodium), oscillated for 5 min by a vortex mixer (MIX-25P, Miou Instrument Co., Ltd., Hangzhou, China), and stored at room temperature.

Animal Operating Methods. All animals received an intragastric administration using a feeding atraumatic needle once per day at 9:00 am for 30 consecutive days. Their body weight and food intake were monitored every week. After 30 min of drug administration, a group of five mice in the exercise group were exposed to the Sleep Interruption (KSYY-I 18S418M1, Yikang Technology Co., Ltd., Beijing, China) and they were made to perform dynamic movements. The principle of sleep interruption was applied with the athletics stimulus. The instrument parameters were set as follows: speed, 20 s/r; number of turns, 3 r; rest time, 1 min, and the total daily duration of the experiment is approximately 360 min. The rotation mode is random (forward or reverse). The mice were forced into constant motion.

At about 4:00 pm, the mice in the same group were dropped into a rectangle glass swimming pool ($50 \times 40 \times 30$ cm³, freshwater depth of 25 cm, and maintained at 25 ± 1 °C) for forced swimming test. In addition, the weights on the tail root of each mouse corresponded to 5% of body weight as evaluated. The swimming time was regarded as the time spent by the mice floating in water with struggle and necessary movements until exhaustion set in. Exhaustion was

determined when there was a loss of movement coordination and failure to return to the surface of the water to breathe within a 5 s period. At the end of the process, mice were dried with a towel and took to their cages to rest, the exercise task was completed. Water in the swimming pool was replaced after each session.

This antifatigue screening model with OCT was established following previous models with slight changes.⁴ In the last 2 days of the experiment, the forelimb grip strength, autonomic activities, and autonomous swimming were completed, and the corresponding fatigue biochemical index was determined (Figure 1).

Autonomic Activities. Autonomic activity is a common indicator of an animal's mobility and is often used to reflect the energy state of mice.²¹ The tests were performed on the 28th day of treatment by OCT, after 30 min of intragastric administration, and the mice were placed in an autonomous chamber in Animal Behavior Analysis System (JLBehv, Shanghai Jiliang Software Technology Co., Ltd., Shanghai, China). After a 5 min adaption time, various behaviors of the mice were observed, and analysis movement videos that lasted 10 min were recorded; the mice movement tracks, including total distance (m), average velocity (mm/s), the number of activities (times), rest time (s), activity time (s), and degree of linearity were also determined. The test was in triplicates.

Forelimb Grip Strength. The forelimb grip strength of mice was tested after treatment with OCT supplementation for 28 days. A Grip tester (YLS-13A, Jinan Yiyuan Science and Technology Development Co., Ltd., Shandong, China) was used to evaluate the forelimb grip strength of mice.² The force transducer equipped with a metal rod (diameter: 2 mm, length: 7.5 cm) was used to gauge the amount of tensile force for mice. As mice gripped the metal rod, its tail was held and pulled backward gently while the two forelimbs gripped the rod, which triggered a "counter pull" causing them to loosen their claws. This maximum grip force of mice was automatically recorded on a digital force transducer. The forelimb grip strengths of all mice were measured on the same day. The test was performed six consecutive times, and the average measure of maximal force (in g) was recorded; this low-force system was used to measure the forelimb grip strength of the mouse forelimb.

Autonomous Swimming Test. Endurance performance is a significant parameter to assess antifatigue, evaluated by autonomous swimming test. A swimming test was conducted referencing the previously reported method with minor modifications.²² On the 30th day of the experiment, 30 min after the oral administration of OCT, the mice were taken into a columnar swimming pool (height 35 cm, radius 10 cm) containing freshwater maintained at 25 ± 1 °C, approximately 25 cm deep. Mice were placed in the pool and their behavior recorded over a 3 min test period. The durations of swims and rests were manually recorded by a trained experimenter during the last period of the test, and rest was defined when the mice ceased struggling and were floating motionlessly on the water, making minimal efforts necessary to keep their head above the water. The endurance performance of the mice was surveyed as the swimming and resting time, recorded from the beginning to exhaustion.

Blood Biochemical Assessments. After 30 days, mice were euthanized with ethyl ether 60 min after the last intragastric administration of OCT. After anesthetization, the whole blood samples were gathered in respective heparinized tubes. The blood collection was done by removing the left eyeball of each mouse without anticoagulation. Serum was promptly prepared by centrifugation at 3000 rpm, at 4 °C for 15 min, and then stored at 4 °C till further analysis. Blood biochemical assessments, blood lactic acid (BLA), lactate dehydrogenase (LDH), superoxide dismutase (SOD), and glutathione peroxidase (GSH-Px) concentrations, were carried out using a commercially available enzyme-linked immunosorbent assay (ELISA) kit (Jiangsu Meimian Industrial Company, Jiangsu, China).

Glycogen Concentration Analysis. After blood collection, liver and hind-limb skeletal muscle tissues of mice were immediately dissected, frozen in liquid nitrogen, and kept at -80 °C until glycogen concentration analysis was performed as described previously.²³ The brief procedure for glycogen concentration analysis is as follows: 100

mg of liver and muscle tissues were cut, weighed, and homogenized in the Tris-HCl buffer. After centrifugation for 15 min at 13 000 rpm and 4 °C, the supernatant solutions were used for further experiments. Tissue extracts (30 μ L) or standard glycogen (Sigma-Aldrich Co., Ltd., St. Louis, MO) were added to wells of a 96-well plate, and 200 μ L of iodine-potassium iodide reagent was added. The 96-well plate was rested for 10 min before the absorbance value was measured at 460 nm in the ELISA reader. The liver glycogen (LG) and muscle glycogen (MG) contents were measured using commercial kits (Meimian Industrial Co., Ltd., Jiangsu, China). All biochemical parameters were determined using a commercial diagnostic kit according to the manufacturer's recommended instructions.

Tissue Sample Preparation. The hind-limb was skinned; the muscle was separated from the tibia and fibula bones of the experimental mice. The skeletal muscle and gastrocnemius were dissected immediately from their surrounding muscles. Muscle tissues were collected and immediately cleaned with ice-cold saline, stored in liquid nitrogen, and then kept at -80 °C for genome-wide gene expression profiling analysis. Then, brains, hearts, livers, lungs, kidneys, and spleens were cut, rinsed in normal phosphate-buffered saline (PBS) solution, and blotted dry. The entire weight and main organ weights relative to the individual's body weight (organ coefficient, %) were recorded and calculated.

Histopathology Analysis. For histological analysis, the midsegments of the gastrocnemius and skeletal muscle tissues were resected, fixed in 4.0% (v/v) paraformaldehyde solution for 24 h, tissue sections of mice were made transparent using dimethylbenzene, and finally, inlaid in molten paraffin. Sections of the gastrocnemius and skeletal muscle tissues were cut into 5 μ m thick slices on a Microtome (Thermo Histostar), put on clean glass slides, and dried overnight at 37 °C. These sections were stained for 6 min and used for the [hematoxylin and eosin (H&E) stained] morphological and pathological evaluation of muscle cells, inspected under a microscope equipped with a digital imaging system (BX-43 + MD-50, Olympus Corporation, Tokyo, Japan).

Gene Expression Profiling Analysis. RNA sequencing was performed at the Shanghai Kangcheng Biological Engineering Co., Ltd. (Shanghai, China). RNA quality was assessed using the Agilent 2100 bioanalyzer and the RNA 6000 Nano Chip (Agilent Technologies, Amstelveen, The Netherlands). RNA samples were quantified using the machine of ND-2000 spectrophotometer (Thermo Fisher Scientific Company, Waltham, MA). Three gastrocnemius RNA samples of mice from each group were stochastically selected. The SuperScript double-strand complementary DNA (ds-cDNA) synthesis kit from Invitrogen Company Ltd. was used to synthesize ds-cDNA from a 5.0 μ g total RNA sample. After precipitation and purification with ethanol under 0 °C, 1.0 μ g of ds-cDNA sample was labeled with the NimbleGen One-Color DNA from the NimbleGen labeling kit (NimbleGen Co., Ltd.) at 37 °C for 2 h under the company's agreement. Then, 0.1 volume of 0.5 mM ethylenediaminetetraacetic acid (EDTA) was used to stop the labeled reaction, and the cDNAs samples were precipitated by isopropanol/ethanol mixed solution. Gene chips were hybridized at 42 °C with 4.0 μ g of labeled cDNA samples in the NimbleGen hybridization buffer plus component A in the hybridization chamber (NimbleGen Co., Ltd.) for 20 h. This step was followed by washing in an ozone-free surrounding, the gene expression chips were scanned by a GenePix 4000B scanner (Axon Co., Ltd.), and signal data of gene chips were collected. Scanned images of gene chips were imported into the NimbleScan software (version 2.5) for grid calibration and gene expression multiple analyses. Expression data of gene chips were normalized via the quantile normalization method, and the robust multichip average (RMA) algorithm was calculated by the NimbleScan. After finishing the normalization step of raw data, the probe level files (*_norm_RMA.pair) and gene-level files (*_RMA.calls) were obtained and imported into the Agilent GeneSpring GX software (version 12.1) for further analysis. Differentially expressed genes of gene chips were identified via fold change (FC) or *T*-test filtering between the three groups. The up- and downregulated expression genes with an absolute FC larger than a selected threshold

and FC lower than a selected threshold were used in the comparison of EX and EX-OCT groups. Hierarchical clustering was performed using MeV 4.9.0 software (<https://sourceforge.net/projects/mev/>).

Total RNA Extraction, Integrity Test, and cDNA Synthesis.

Gastrocnemius tissue (100 mg/sample) was resected and grounded in the liquid nitrogen mortar. Total RNA was extracted with the TransZol-Up reagent from TransGen Biotech Company (TransGen Biotech Co., Ltd., Beijing, China) and the steps were in accordance with the operation manual of the manufacturer. Extracts were immediately stored in a -80°C refrigerator. The quality, purity, and concentration of total RNA were determined using a Nano-Drop 2000c ultramicro spectrophotometer (Gene Co., Ltd., Beijing, China). The RNA integrity of samples was checked by a standard denaturing agarose gel electrophoresis experiment (Bio-Rad Co., Ltd.). High-quality RNA samples were chosen for the next step experiments. About 2.0 μg of total RNA samples were used as a template to transcribe first-strand cDNA synthesis for RT-qPCR experiments (cDNA reverse transcription kits, Applied Biosystems Co., Ltd.).

Real-Time Quantitative PCR. Real-time quantitative polymerase chain reaction (RT-qPCR) was performed using the CFX96 Real Time-PCR system (SYBR Select Master Mix protocol) (Applied Biosystems Co., Ltd.). The focus was on peripheral myelin protein 22 (Pmp22), Trim63, unc-51-like kinase 3 (Ulk3), arrestin domain containing 2 (Arrdc2), Prx, B cell leukemia/lymphoma 3 (Bcl3), glycoprotein hormones, α subunit (Cga), synaptic vesicle glycoprotein 2 a (Sv2a), Cacna1h, Mybpc3, and β -actin (control for RT-qPCR) of antifatigue related genes. The PCR conditions were as follows: initial denaturation for 3 min at 94°C ; denaturation for 30 s at 94°C ; annealing for 40 s at 60°C ; extension for 1 min at 72°C ; and the cycle of 42 times. Gene expression levels of the target genes were computed based on the $2^{-\Delta\Delta C_t}$ (RQ) formula. The specificity of PCRs was analyzed by the dissociation curve and 2.0% agarose gel electrophoresis (Bio-Rad Co., Ltd.). Primers of genes were designed using Primer Premier 6.0 software (Premier Ltd., Canada), and are listed in Table 1. All PCR products were within the linear range of the

(Servicebio Co., Ltd., Cat. no. G2002, Wuhan, China), mixed with 4 μL of cocktail, 4 μL of phosphatase inhibitors, and 4 μL of phenylmethylsulfonyl fluoride (PMSF) at 4°C for 1 h. The suspension solutions were centrifuged at 13 200 rpm for 15 min at 4°C ; then, the supernatant solution was pipetted and protein concentrations were detected with the bicinchoninic acid (BCA) protein assay kit (Beyotime Biotechnology Co., Ltd., Shanghai, China) in accordance with the company's instructions. The prepared protein samples (200 μL) were fully mixed with appropriate amounts of 5 $\times$ sodium dodecyl sulfate (SDS) loading buffer [10% 2-hydroxy-1-ethanethiol, 20% glycerol, 4% SDS, 0.3% bromophenol blue, and 125 mM Tris-HCl (pH = 6.8)] heated for 10 min at 95°C and then stored at -20°C .

Western Blot Analysis. About 7–8 μL of protein samples were denatured by heating and separated through electrophoresis onto a 10% SDS-polyacrylamide gel electrophoresis (PAGE) gel, and transferred to a poly(vinylidene difluoride) (PVDF) membrane. The membranes hosting the target protein were blocked using 5% bovine serum albumin (BSA) (2 g of BSA power in 40 mL of 1 $\times$ Tris-buffered saline with Tween 20 (TBS-T)) at 25°C for 1 h. After being blocked, the membranes were incubated with the corresponding primary antibody (1:1000 dilution in 5% BSA with TBS-T) at 4°C for 8–10 h. These were washed thrice (10 min per time) afterward with TBS-T at 25°C and re-incubated in a secondary antibody of antimouse and/or antirabbit IgG HRP (1:5000–10 000) conjugate at 25°C for 1 h. The membranes were washed thrice (10 min) again with TBS-T. Immunoreactive proteins on the membranes were detected using the ECL Plus Western blot detection system (no. 721BR10883, Pierce, Rockford, IL), and the signals of membranes were recorded in the Gel Imaging system (Chemi Doc XRS+, Bio-Rad Co., Ltd.). The relative target protein expression compared to the control protein was determined by calculating the integral optical density value of each band using the Gel Imaging System.

Statistical Analysis. Each experiment was repeated three times, and the related data were analyzed by SPSS software (version 2.5, SPSS, Chicago, IL). Grip results were shown as the mean $\pm$ standard error mean (SEM), and other results were shown as the mean $\pm$ standard deviation (SD) and were analyzed using one-way analysis of variance (ANOVA) test, followed by Tukey's test, when comparisons proceed intergroups and indicated by vertical bars in figures. Percentage comparisons were analyzed by utilizing χ^2 test. The probability value p -value less than 0.05 was considered statistically significant.

Table 1. Primers of Genes for RT-qPCR

gene name	sense (5'–3')	anti-sense (5'–3')
Pmp22	TTG CTG GTC TGT GCG TGA TGA GT	GCG AAG CCA TAG GAG TAG TCA GTG T
Trim63	ATC ACG CAG GAG CAG GAG GAG A	CTT GGC ACT TGA GAG GAA GGT AGC
Ulk3	AGG AAG GAG GCA ATT AAG GCG AAG G	CGA AGC AGC TCT TGG ACA GTT GTG
Arrdc2	GAG AAG GTG GCA CGA TCC TGG TA	ATG ACC TCG CCT GGA GTG TAG C
Prx	AGG CGG CGG CAA AGA AGG AAT C	TCT CAA AGA ACA CAC GGG CAC TCA
Bcl3	GCA GCA GTC GTC TCA GCT CCA A	GGC AGG AAG GCA GGT GTA GAT GTT
Cga	GGT CCA AGA AGA CAA TGC TGG TTC C	GGC ACT CCG TAT GAT TCT CCA CTC T
Sv2a	GGA CAA GAT TGG CAG GCT CAG AAT G	GCG TTC CAG GAT GCG ATG CTA AC
Cacna1h	GCC ATC GTC CTC CTG TCC ATC AT	GCC ATC TTC AGT AGC TTC AGC ACA C
Mybpc3	TGC CTG GTG TGA CTG TTC TCA AGA T	CCT CCG CTG ACC TTG GCT TCT T
β -actin	GTG AAA AGA TGA CCC AGA TCA TGT	AGG ATC TTC ATG AGG TAG TCT GTC

exponential phase of PCR amplification. Then, the relative expression levels of target genes were calculated using the Applied Biosystems software (Applied Biosystems Co., Ltd.), and the expression ratio of the target gene to that of β -actin is represented in bars.

Total Protein Extraction. Gastrocnemius tissues (100 mg/sample) were fragmented in liquid nitrogen precooled mortar; the homogenized samples were then poured in 400 μL of RIPA buffer

RESULTS

Effect of OCT on the Body Weight, Food Intake, and Organ Weight. The body weights, food intake, and main organs (including brain, heart, lungs, liver, kidneys, and spleen) of mice involved in 30 days forced exercise experiment were measured and recorded. In addition to normal growth, there were no significant differences in the curve of weights and food intake in the three groups of mice ($p > 0.05$) (Figure 2A,B). The changes in the main organ weights were expressed as an organ/body weight percentage. The results suggest that continuous OCT supplementation did not influence the relative tissue weight when compared with the CON group ($p > 0.05$) (Figure 2C).

Effect of OCT on Energy and Endurance Performance. After 4 weeks of OCT supplementation, the autonomic activities of mice were analyzed. We observed that the total distance ($p < 0.05$), average velocity ($p < 0.05$), number of activities ($p < 0.01$), activity time ($p < 0.01$), and degree of linearity ($p < 0.05$) of the EX-OCT group significantly increased compared to the EX group within the same recording time (Figure 3), but the rest time ($p < 0.05$) decreased (Figure 3D).

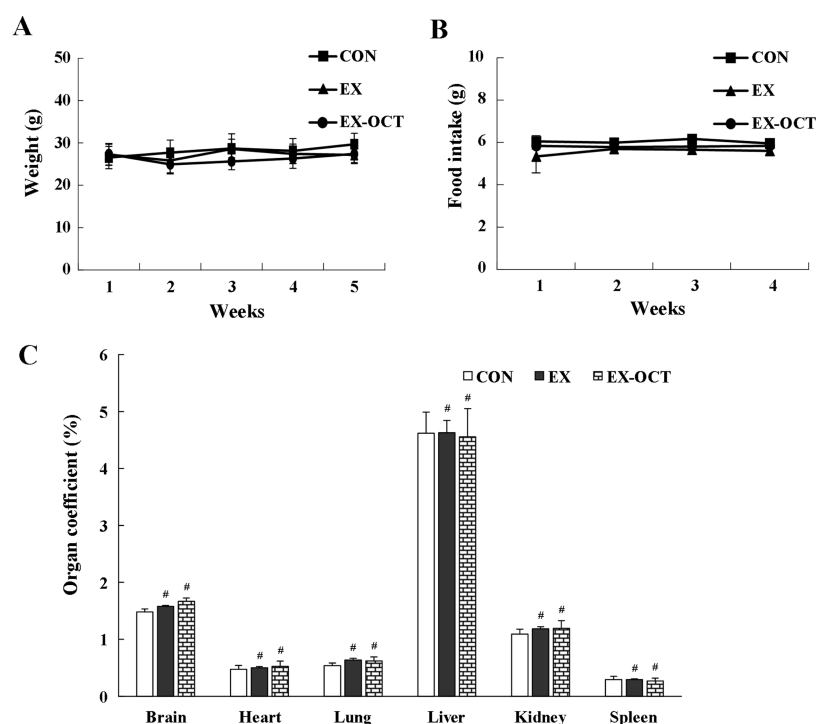


Figure 2. Effects of octacosanol on the weights. Bodyweight (A), food intake (B), and relative weights of main organs (C), including brain, heart, lungs, liver, kidneys, and spleen of C57BL/6 mice (organ coefficient (%) = organ/body weight $\times$ 100). CON vs EX or EX vs EX-OCT; the values represent mean $\pm$ SD. ** p < 0.01, * p < 0.05, and # p > 0.05.

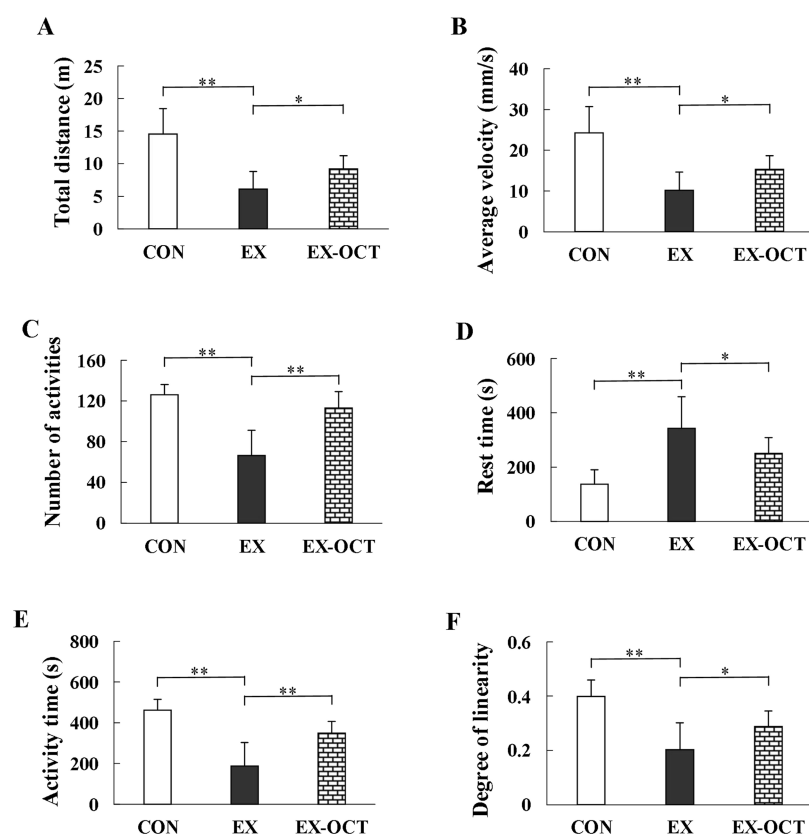


Figure 3. Effects of octacosanol on the autonomic activity for energy performance. Total distance (A), average velocity (B), number of activities (C), rest time (D), activity time (E), and degree of linearity (F) in C57BL/6 mice (n = 10). The values represent mean $\pm$ SD. ** p < 0.01, * p < 0.05, and # p > 0.05.

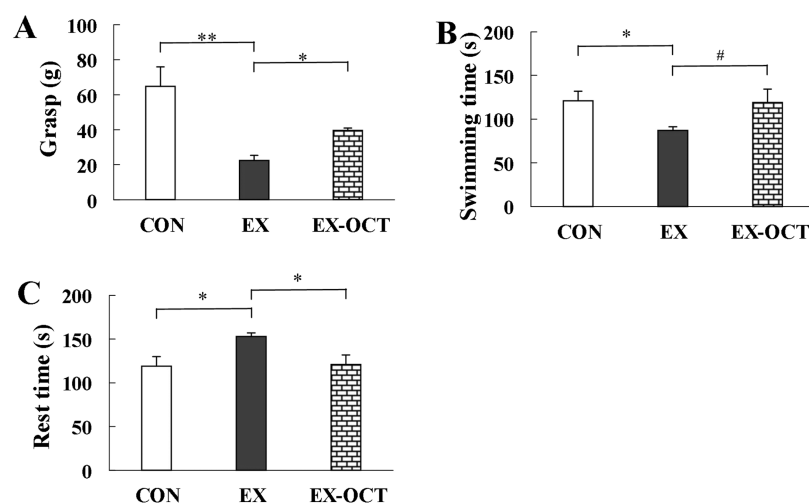


Figure 4. Effects of octacosanol on endurance performance. Forelimb grip strength (A), autonomous swimming time (B), and rest time (C) in C57BL/6 mice ($n = 10$). The values represent mean $\pm$ SEM/SD. ** $p < 0.01$, * $p < 0.05$, and # $p > 0.05$.

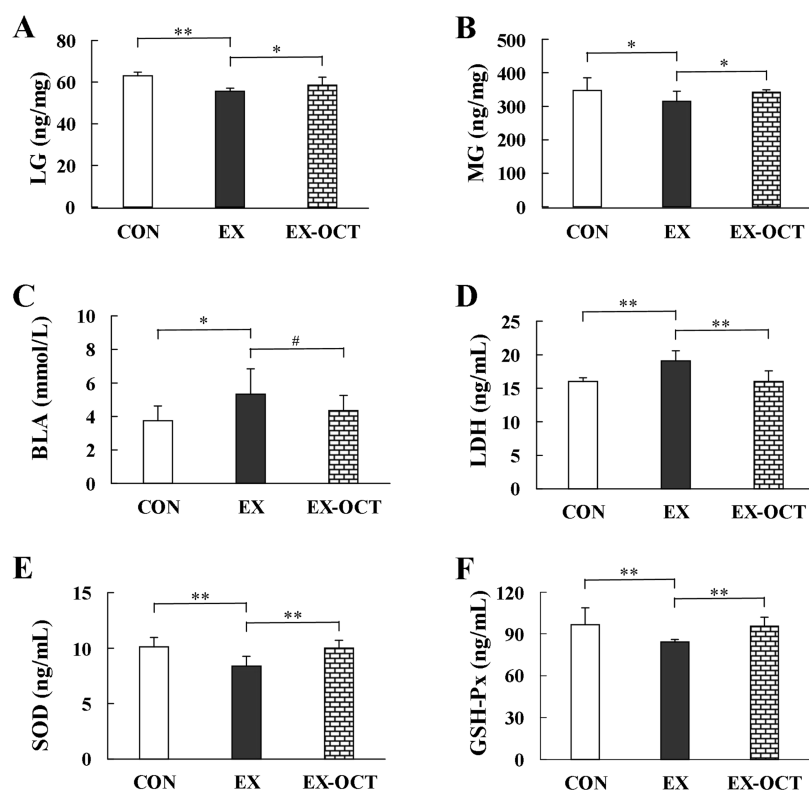


Figure 5. Effects of octacosanol on the assessments of the biochemical index. LG (A), MG (B), BLA (C), LDH (D), SOD (E), and GSH-Px (F) levels in the liver, hind-limb skeletal muscle tissues, and whole blood of C57BL/6 mice ($n = 5$). The values represent mean $\pm$ SD. ** $p < 0.01$, * $p < 0.05$, and # $p > 0.05$.

The grip of the forelimb is a measure of endurance in mice.² The forelimb grip strength was observably higher in the EX-OCT mouse group than the EX mouse group ($p < 0.05$) (Figure 4A). The results showed that exercise model building is successful, and the grip was evidently different in the CON and EX groups ($p < 0.01$) (Figure 4A). The grips of mice in CON, EX, and EX-OCT groups are 64.79 ± 11.11 , 22.48 ± 2.77 , and 39.63 ± 1.38 g, respectively. After prolonged exhausting exercise, the mice's grip reduced by 2.88-fold. Interestingly, mice with OCT supplementation had a 1.76-fold increase in grip.

In addition, swimming time is an important indicator for exercise endurance and is evaluated by antifatigue compounds.²⁴ The autonomous swimming and rest times of mice were measured to evaluate the inhibitory effect of OCT on fatigue. We observed that the swimming capacity increased after administration of OCT (200 mg/kg, $p < 0.05$) for 29 days compared with that of the EX group (Figure 4B). In a span of 3 min, the swimming times of mice in CON, EX, and EX-OCT groups were 121.1 ± 34.64 , 87.1 ± 13.71 , and 119.1 ± 48.39 s, respectively, indicating that the mice had a 1.37-fold increase in swimming time after the addition of OCT. On the contrary,

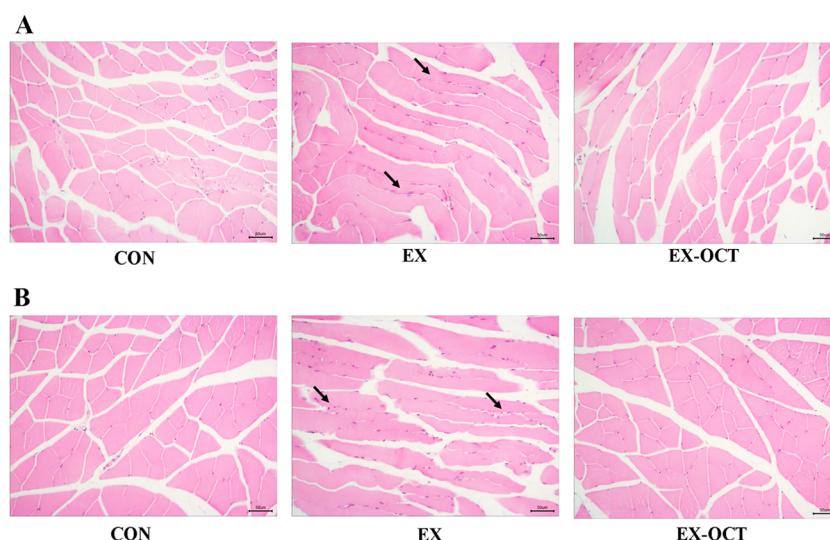


Figure 6. Effects of octacosanol supplementation on morphology. Gastrocnemius tissues (A) and skeletal muscle tissues (B) of C57BL/6 mice. The arrows indicate that the local muscle fiber structure is blurred or dissipated, loose, slight cracks exist, and twisted. Specimens were photographed with a biological microscope equipped with a digital imaging system (BX-43 + MD-50), H&E stain, magnification: 10×20 , and scale bar, $50 \mu\text{m}$.

the rest time after swimming was observably decreased ($p < 0.01$) (Figure 4C). The rest time of mice after exercise reduced by 1.29-fold than that in the CON group; furthermore, the mice group supplemented with OCT (200 mg/kg for 29 days) had increased rest time by 1.26-fold. These results suggest that OCT actively affects the energy and endurance of mice. To further clarify its potential mechanism, the biochemical indicator related to fatigue were appraised.

Effect of OCT on Glycogen Concentration. Energy consumption and deficiency affect the energy supply and maximal power output in physical fatigue during exercise.^{23,25} Stored glycogen in muscle and liver tissues is the primary source of energy during exercise. In this study, the LG in the EX group decreased by 11.82% from 63.03 ± 1.79 to 55.58 ± 1.67 ng/mg ($p < 0.01$) compared with the CON group. OCT treatment slightly ameliorated exercise-induced LG loss, which was increased to 58.48 ± 3.81 ng/mg ($p < 0.05$, Figure 5A). Moreover, MG followed the same change trend as LG, and the MG contents of mice in CON, EX, and EX-OCT groups were 347.89 ± 39.29 , 315.44 ± 29.77 , and 341.65 ± 7.53 ng/mg, respectively. The MG content in the OCT supplementation group is increased by 8.31% ($p < 0.05$, Figure 5B) compared with the EX group. These results suggest that OCT can slightly improve the decrease of LG and MG contents of mice induced by forced exercise and then enhance the exercise endurance of mice.

Effect of OCT on Biochemical Assessments. Activities or levels of certain enzymes and biochemical parameters can forthrightly reflect the true status of physical fatigue. They are widely used to estimate the antifatigue activity of test substances.²⁶ As shown in Figure 5, exposure to the forced exercise led to an enhancement in BLA and LDH levels in comparison to the CON group. The BLA increased the most after exercise, increased by 11.82% from 3.74 ± 0.88 to 5.33 ± 2.43 mmol/L ($p < 0.05$). OCT reduced the concentration of BLA to 4.35 ± 0.90 mmol/L ($p > 0.05$, Figure 5C), although the change was not significant. Also, the LDH was increased by 18.99% ($p < 0.01$) after exercise in mice; OCT treatment observably reduced the LDH content in the blood of mice,

from 19.06 ± 1.52 to 16.03 ± 1.57 ng/mL ($p < 0.01$, Figure 5D). When compared with the CON group, exposure to the forced exercise test led to a reduction in SOD and GSH-Px levels, from 10.12 ± 0.84 and 96.68 ± 12.02 ng/mL to 8.39 ± 0.81 ($p < 0.01$, Figure 5E) and 84.23 ± 1.80 ng/mL ($p < 0.01$, Figure 5F), respectively. Interestingly, OCT inhibited the above changes, with the SOD and GSH-Px levels increasing to 10.01 ± 0.70 ($p < 0.01$, Figure 5E) and 95.51 ± 6.55 ng/mL ($p < 0.01$, Figure 5F), respectively. Based on these collective effects of OCT on BLA, LDH, SOD, and GSH-Px, we conclude that OCT has antifatigue activity in mice, and 200 mg/kg is a potential dose for the development of OCT as an antifatigue drug and/or functional food. Further study is warranted to elucidate its antifatigue-related gene regulation.

Effect of OCT on Histopathology of Muscle. Muscle mass is the main determinant of muscle strength and strength and is closely related to an organism's exercise endurance capacity.²⁷ Histopathological examination of the gastrocnemius and skeletal muscle did not reveal significant lesions between the CON, EX, and EX-OCT groups (Figure 6). The results indicated that no clinical signs of tissue-specific toxicity were observed after OCT treatments. Nevertheless, all cell nuclei were situated at the edge of the muscle fibers. In contrast, the skeletal muscle fibers in the forced exercise mice showed typical features of striated-muscle degeneration, such as blurred or dissipated (Figure 6B, as shown in the arrow). Furthermore, local muscle fiber structures became loose, with slight cracks and twists, and the cell nuclei were more mixed and distributed in a disorderly manner (Figure 6B, as shown by the arrow). Likewise, the gastrocnemius muscle of mice also showed these characteristics (Figure 6A). After treatment with OCT for 30 days, the anomalies were improved to a large extent.

OCT Modulates the Expressions of Factors in Gastrocnemius Tissues. To elucidate the possible mechanism for the antifatigue effect of OCT, a cDNA microarray was implemented to obtain gene expression profiles of gastrocnemius tissues. The experiment data, exhibited in the investigation including genes of EX and EX-OCT groups, showed increased and decreased expression levels of more than

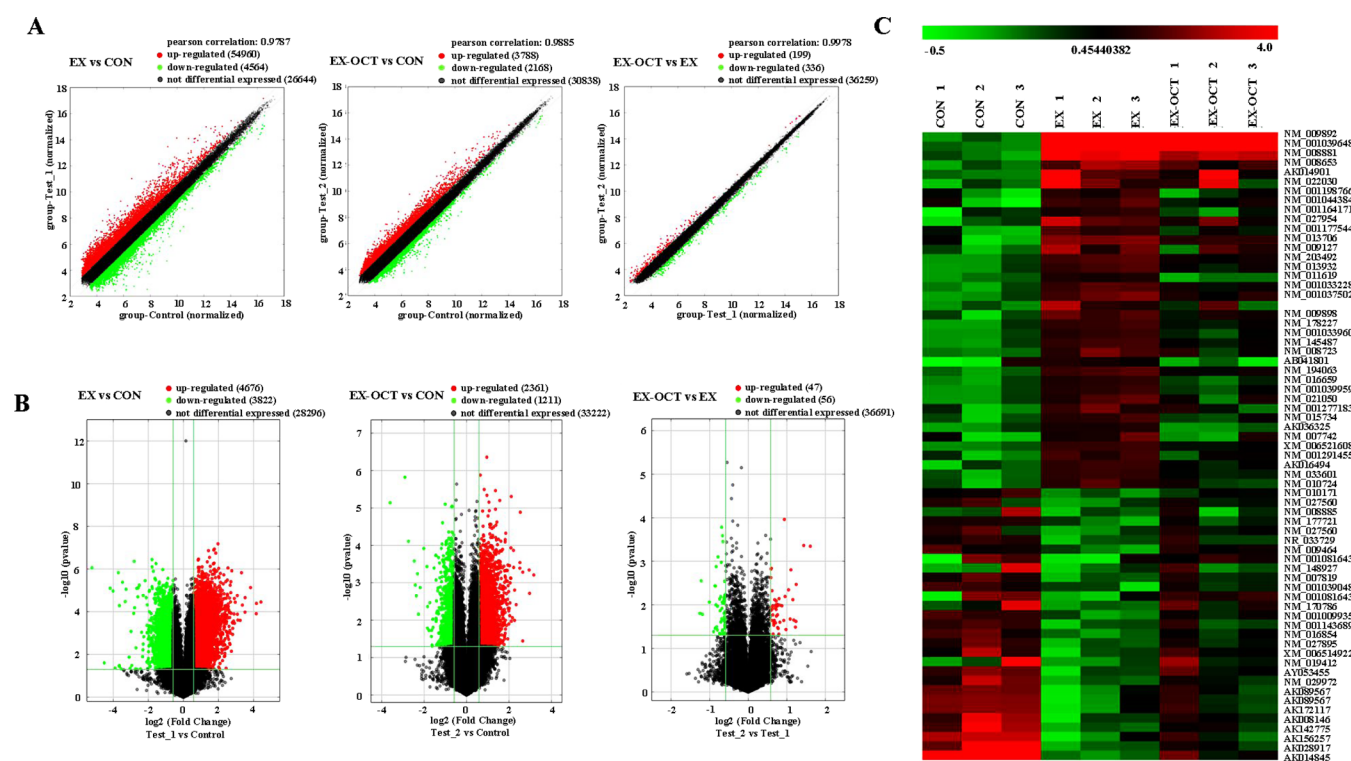


Figure 7. Genome-wide gene expression profiling analysis of gastrocnemius tissues from C57BL/6 mice. The scatter plots (A) and volcano plot (B) analysis of differentially expressed genes in mice, EX vs CON, EX-OCT vs CON, and EX-OCT vs EX. $p < 0.05$; fold change value is greater than 1.5, and/or fold change value is less than 0.75. Genes that were relative to exercise fatigue were labeled. Heat map (C) of differentially expressed genes related to fatigue affected by octacosanol in mice for EX and EX-OCT groups compared with the CON group. The hierarchical clustering was implemented using the program MeV 4.9.0.

1.5-fold or less than 0.75-fold compared with the CON group. The data of gene chip (36 894 signals) was filtered and analyzed by hierarchical clustering. 38 decreased expression genes and 29 increased expression genes were discovered to be administered by OCT treatments (Figure 7). The list of gene names and FC are as listed in Tables 2 and 3.

OCT Affects the Expressions of Fatigue-Related Target Genes. To explore the molecular mechanism of OCT on antifatigue, the related target genes were analyzed differentially using the β -actin expression value as control using RT-qPCR analysis. It was observed that messenger RNA (mRNA) expressions of gastrocnemius tissue genes, including Pmp22, Trim63, Ulk3, Arrdc2, Prx, Bcl3, Cga, Sv2a, Cacna1h, and Mybpc3 were differentially expressed with the treatment of forced exercise based on the CON group. OCT supplementation significantly regulated the expressions of these genes when compared to the EX group (Figure 8A,B). Among these genes, increased expression genes were Pmp22, Trim63, Ulk3, Arrdc2, and Prx at 1.27, 1.45, 1.50, 1.63, and 1.88-fold in gastrocnemius tissues, respectively, while experimental data from RT-qPCR were 2.42, 4.69, 4.23, 5.56, and 5.39-fold, respectively (Figure 8C). For decreased expression genes, Bcl3, Cga, Sv2a, Cacna1h, and Mybpc3 were selected. They were downregulated by OCT treatments at 1.53, 1.57, 2.13, 2.78, and 3.21-fold by gene chip analysis, respectively, and RT-qPCR analysis indicated that expression levels of those genes were 1.71, 1.90, 1.46, 2.34, and 4.41-fold, respectively (Figure 8E). All of the above comparisons were conducted between the EX-OCT group and the EX group. For those increased expression genes, the Pearson correlation coefficient of gene microarray analysis and RT-qPCR was 0.6962 ($p < 0.01$)

(Figure 8D). For those decreased expression genes, the Pearson correlation coefficient was 0.6501 ($p < 0.01$) (Figure 8F). The results of gene chip analysis were reliable and over 67 genes took part in the antifatigue action of OCT treatments. It means that OCT supplementation alleviated fatigue effects via modulating related gene expressions.

Functional Validation of Identified Genes through Gene Ontology (GO) Analyses. Gene ontology (GO) analysis mainly includes biological processes, cellular components, and molecular functions. To explore the functional relevance of the identified genes related to fatigue, we performed GO analyses using online software of g:Profiler (<https://biit.cs.ut.ee/gprofiler/gost>), and performed GO analysis for all of the 67 genes and found 8 GO terms (Figure 9), which were related to myofibril, contractile fiber, calcium-dependent adenosinetriphosphatase (ATPase) activity, and regulation of the multicellular organismal process. The interaction network of genes and GO function was finally analyzed using Cytoscape 2.8.2 software (<http://www.cytoscape.org/>) (Figure 9A).

Main Network among Genes Affected by OCT. We further uploaded the normalized gene chip data into an online software STRING (<https://string-db.org/cgi/>). The STRING database showed the functional gene association network (Figure 10). Forty-three genes with at least one interaction formed the main network among genes affected by OCT, indicating that OCT played a role in relieving fatigue through holistic network regulation on these genes.

OCT Affects the Expressions of Fatigue-Related Target Proteins. Furthermore, to determine whether the activation of fatigue-related genes was changed by OCT in

Table 2. Genes Downregulated by Octacosanol from Microarray Analysis

gene ID	gene name	EX/CON fold change	EX/CON <i>p</i> value	EX/EX-OCT fold change	EX/EX-OCT <i>p</i> value
NM_007742	Col1a1	2.12	0.0429	1.68	0.0012
NM_001164171	Myh6	4.97	0.0466	4.37	0.0544
NM_015734	Col5a1	2.15	0.0458	1.65	0.0529
NM_008881	Plxna1	8.90	0.0093	2.26	0.0251
NM_001177544	Hist1h2ao	3.50	0.0430	2.50	0.0612
NM_033601	Bcl3	2.05	0.0464	1.53	0.0382
NM_001198766	Postn	5.61	0.0423	4.68	0.0122
NM_001291455	Dpysl3	2.06	0.0470	1.47	0.0880
NM_011619	Tnnt2	2.82	0.0452	2.95	0.0800
NM_007742	Psmb8	2.01	0.0475	1.46	0.0974
NM_009892	Chil3	14.08	0.0408	2.01	0.0225
NM_001039648	Gm20767	10.84	0.0400	1.51	0.0300
NM_013706	Cd52	3.27	0.0399	1.47	0.0449
NM_009898	Coro1a	2.66	0.0457	1.56	0.0565
AK036325	Syn3	2.13	0.0418	2.18	0.0494
NM_001277183	Gm1966	2.16	0.0470	1.44	0.0485
NM_001039959	Ahnak	2.28	0.0420	1.81	0.0334
NM_001033960	Rabgap1	2.40	0.0409	1.88	0.0271
NM_016659	Klra1	2.29	0.0466	1.98	0.0478
NM_008723	Npm3	2.37	0.0429	1.42	0.0436
NM_013932	Ddx25	2.98	0.0394	1.94	0.0412
NM_178227	Scn3b	2.54	0.0389	1.79	0.0445
NM_145487	Prr3	2.38	0.0367	1.62	0.0493
NM_203492	Mrgprg	3.03	0.0357	1.79	0.0464
AK016494	4931431B13Rik	2.06	0.0355	1.47	0.0494
NM_021050	Cftr	2.16	0.0311	1.40	0.0646
NM_001037502	Defb28	2.70	0.0297	1.41	0.0525
NM_001033228	Itga1	2.71	0.0260	1.65	0.0459
NM_194063	Rhox3a	2.33	0.0236	1.44	0.0575
XM_006521608	Gm5215	2.07	0.0197	1.46	0.0478
NM_009127	Scd1	3.04	0.0192	1.57	0.0413
NM_027954	Syce2	3.71	0.0115	1.64	0.0419
AK014901	4921515L22Rik	6.76	0.0089	2.03	0.0307
	Cga	2.66	0.0017	1.57	0.0418
AB041801	Cacna1h	2.34	0.0494	2.78	0.0435
NM_022030	Sv2a	5.62	0.0732	2.13	0.0481
NM_001044384	Timp1	5.36	0.04424	3.44	0.0531
NM_008653	Mybpc3	7.44	0.04632	3.21	0.0470

gastrocnemius tissues of mice, related proteins expressions were examined by western blot analysis. In Figure 11, after forced exercise, the amounts of Prx and Trim63 in the gastrocnemius tissues decreased markedly but increased in the OCT treatment group (Figure 11A). Furthermore, the expression of Mybpc3 was also altered, and the expression levels of Mybpc3 increased with exercise and reduced with OCT treatment. These results were consistent with microarray and RT-qPCR analysis results, although the expression changes of Cacna1h at the protein level were less obvious (Figure 11A).

DISCUSSION

Fatigue can be defined as an exercise-induced inability to perform an expected or desired work output,²⁴ which could result in a broad range of physical and mental unfitness, such as inattention, drowsiness, and weakness, and it provokes various disorders that could be nervous, endocrine, metabolic, or immune dysfunction, etc.⁵ Previous studies, including animal experiments, have confirmed that OCT is able to play an antifatigue role.^{11,17,28} The current study demonstrates an antifatigue activity of OCT in the exercise model and on

several markers for fatigue. Especially, it is shown that the body weight, food intake, and main organ relative weight were not markedly changed, and OCT exhibited high safety profiles. OCT supplement could alleviate fatigue symptoms in mice, caused by forced exercise, which includes enhanced forelimb grip strength and swimming endurance.¹¹ OCT has a certain alleviating effect on muscle injury induced by over-exercising in mice. Under normal circumstances, the myofilament is in a contraction state and the arrangement was in perfect order and neat.²⁹ OCT can relieve the slightly blurred or dissipated muscle fibers caused by forced exercise.

Moreover, the role of OCT in the mice is the attenuation of the effects of forced exercise induced on the physiological indicators for fatigue. Previous findings suggest that recovery from exercise-induced fatigue requires repair of the injuries that has already occurred in the body and/or prompts the elimination of metabolites that accumulate during exercise.⁸ In this study, OCT restored the levels of glycogen, BLA, LDH, SOD, and GSH-Px to baseline. The energy for exercise is initially generated from the breakdown of MG, and later stages will be derived from the liver.^{23,25} LG depletion is an important

Table 3. Genes Upregulated by Octacosanol from Microarray Analysis

gene ID	gene name	EX/CON fold change	EX/CON <i>p</i> value	EX/EX-OCT fold change	EX/EX-OCT <i>p</i> value
NM_010171	F3	0.51	0.0441	0.73	0.0487
NM_027560	Arrdc2	0.51	0.0498	0.66	0.0313
NM_008885	Pmp22	0.51	0.0466	0.75	0.0470
NM_177721	Ranbp6	0.50	0.0167	0.74	0.0432
NM_027560	Arrdc2	0.49	0.0453	0.62	0.0405
NR_033729	Fbxl12os	0.49	0.0061	0.64	0.0473
NM_009464	Ucp3	0.48	0.0278	0.56	0.0470
NM_001081643	Xlr3b	0.48	0.0487	0.53	0.0418
NM_148927	Plekha4	0.48	0.0462	0.65	0.0462
NM_007819	Cyp3a13	0.45	0.0470	0.65	0.0436
NM_001039048	Trim63	0.44	0.0023	0.69	0.0481
NM_001081643	Xlr3b	0.44	0.0474	0.45	0.0124
NM_170786	Cntf	0.43	0.0410	0.52	0.0496
NM_001009935	Txnip	0.42	0.0008	0.65	0.0458
NM_001143689	H2-Q4	0.42	0.0046	0.66	0.0477
NM_016854	Ppp1r3c	0.41	0.0376	0.72	0.0440
NM_027895	Ulk3	0.41	0.0158	0.67	0.0282
XM_006514922		0.40	0.0449	0.55	0.0408
NM_019412	Prx	0.37	0.0493	0.53	0.0413
AY053455	Hsd3b2	0.36	0.0031	0.52	0.0466
NM_029972	Ernm	0.33	0.0311	0.68	0.0425
AK089567		0.33	0.0055	0.62	0.0462
AK089567		0.31	0.0056	0.63	0.0484
AK172117		0.31	0.0041	0.61	0.0451
AK008146	2010007E15Rik	0.29	0.0207	0.63	0.0415
AK142775	Gm10428	0.26	0.0327	0.66	0.0442
AK156257		0.25	0.0016	0.63	0.0513
AK028917	Gm4402	0.15	0.0435	0.66	0.0416
AK014845		0.09	0.0117	0.50	0.0496

factor in the development of fatigue; increased glycogen storage and reduced glycogen consumption contribute to enhanced endurance and locomotors capacity.³⁰ However, in this study, the MG level did not increase as much as LG. It might be because OCT promoted glycogenolysis restraint and/or gluconeogenesis. Generally, the muscle in high-intensity exercise can produce plenty of LA almost simultaneously as it obtains enough energy from anaerobic glycolysis.²⁵ Excess LA is generated and accumulated in muscles, generating fatigue.²⁶ Studies have shown that swimming to exhaustion results in significantly elevated BLA levels.³¹ LDH is an enzyme that catalyzes the reciprocal conversion of pyruvate and lactic acid, which plays an important role in eliminating LA and providing energy under hypoxia.³² Our data showed that the LDH content in serum decreased in EX-OCT groups, indicating that OCT could decrease muscle damage during forced exercise. The result indicated that reduction of BLA and LDH levels may be another pathway through which OCT elicits its antifatigue effect. Fatigue is also related to immune abnormalities in the oxidative stress pathways or multiple inflammatory pathways.¹ Oxidative stress caused by over-exercise leads to the production of excess free radicals contributing intensively to fatigue.³³ The intracellular defense systems including SOD and GSH-Px can scavenge free radicals and their metabolites, and this helps them fight against fatigue.²⁵ Data showed that OCT increased SOD and GSH-Px activities in mice, indicating that OCT can protect against oxidative stress induced by exhaustive exercise, alluding to OCT having an antifatigue effect. OCT has been used extensively as an adaptogen whose properties are the ability

to increase as a nonspecific body to resist fatigue.¹² Although a number of biochemical mechanisms are suggested as modulating these effects, further research is needed to elucidate the nature of OCT effects on fatigue, and particularly, on the metabolite changes that are the ultimate result of genetic changes.

Fatigue may be attributed to a deviation of a series of gene expressions. To investigate the potential molecular mechanism of the antifatigue effect of OCT, a microarray assay was executed using the RNA from the gastrocnemius tissues of mice. From the results of the analysis, validated by RT-qPCR, expression modifications of more than 67 genes were noticed. GO analyses showed that these genes enriched functions in the eight terms related to myofibril, contractile fiber, calcium-dependent ATPase activity, regulation of the multicellular organismal process, etc. From these terms, further studies on the interactive relationship of these genes during the OCT antifatigue modulation process are required. Among these genes, we found several interesting genes or gene families, which may be the target of OCT affecting the antifatigue process. Microarray data would promote investigation into the signal transduction pathways, the gene expressions regulated by OCT, and provide an explanation for the possible molecular mechanism of OCT in the antifatigue effect. OCT regulated the expression levels of fatigue-related genes in forced exercise stimulated in the gastrocnemius tissue of mice. It can be clearly discovered that the expression levels of mRNA including Pmp22, Trim63, Ulk3, Arrdc2, Prx, Bcl3, Cga, Sv2a, Cacna1h, and Mybpc3 were significantly different; the protein expression

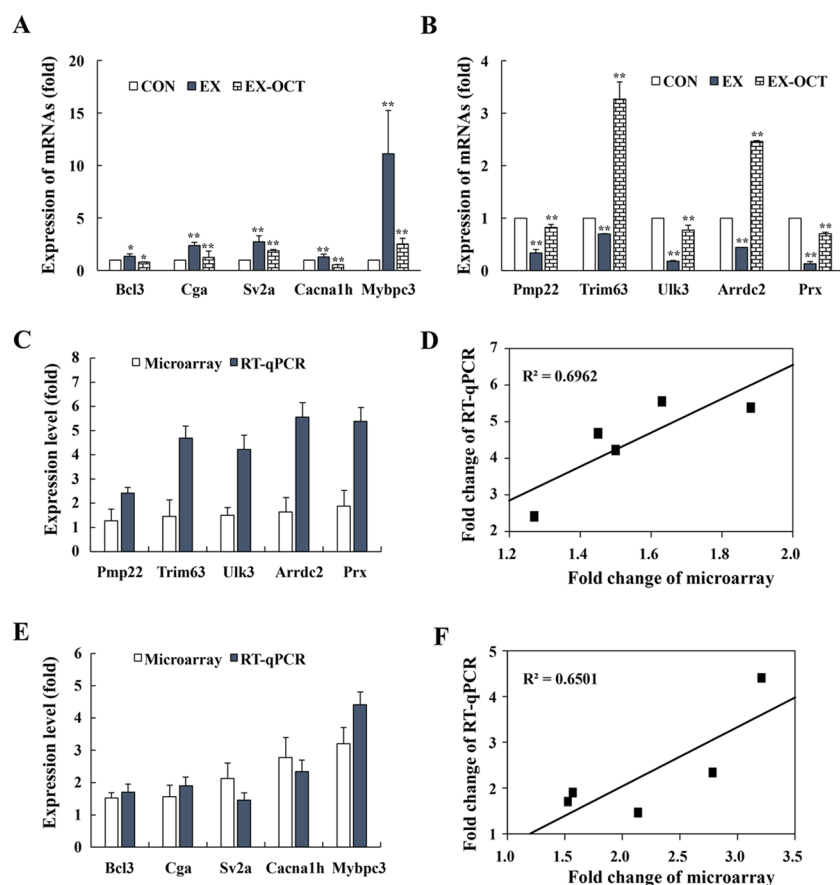


Figure 8. Octacosanol regulated the expression levels of fatigue-related genes in forced exercise stimulated in the gastrocnemius tissue of C57BL/6 mice. Mice treated with octacosanol (200 mg/(kg day)) for 30 days. Total RNA was extracted from the gastrocnemius tissue of mice after octacosanol treatments; comparisons of expression fold change between microarray and RT-qPCR ($n = 3$). These genes included Pmp22, Trim63, Ulk3, Arrdc2, Prx, Bcl3, Cga, Sv2a, Cacna1h, and Mybpc3, which were differentially expressed, caused by forced exercise fatigue in the gastrocnemius tissue of C57BL/6 mice. Expression levels of five upregulated genes by the RT-qPCR (A), expression levels of five downregulated genes by the RT-qPCR (B), expression fold change of five upregulated genes (C), linear correlation of five upregulated genes (D), expression fold change of five downregulated genes (E), and linear correlation of five downregulated genes (F). The images shown here are representative data from three independent experiments. Values represent mean $\pm$ SD. ** $p < 0.01$, * $p < 0.05$, and # $p > 0.05$.

levels of Prx, Trim63, Cacna1h, and Mybpc3 were also changed by OCT treatment.

B cell leukemia/lymphoma 3 (Bcl3) is a member of the I κ B family and contributes to regulating transcriptional activation of nuclear factor κ -B (NF- κ B) target genes.³⁴ Unaccustomed strenuous exercise can lead to muscle strength loss, inflammation, and delayed-onset muscle soreness.³⁵ Bcl3 is an NF- κ B transcriptional activator required for muscle atrophy and binds to the promoters of several genes whose collective function describes muscle wasting.³⁶ Studies showed that knockout of the Bcl3 gene in the soleus muscle caused inhibition of fiber atrophy and abolition of NF- κ B activity. The NF- κ B and Bcl3 play a critical role in muscle atrophy.^{37,38} Classical NF- κ B signaling is a key factor in controlling inflammation and it has been found that OCT exerts anti-inflammatory properties through the blocking of mitogen-activated protein kinase (MAPK)–NF- κ B/AP-1 signaling pathway.²⁰ Furthermore, Bcl3 controls Toll-like receptors (TLRs)-induced MAPK activity.³⁹ Besides, NF- κ B can mediate the protein loss induced by tumor necrosis factor- α (TNF- α) in differentiated skeletal muscle myotubes.⁴⁰ Also Bcl3 directs the transcription networks of proteolysis and energy metabolism.³⁶ Our study found that OCT reduces the expressions of Bcl3 in gastrocnemius (Figure 8B), which

inhibits skeletal muscle atrophy, inflammatory transcription factors, and energy metabolism.

Tripartite motif-containing 63 (Trim63, former MuRF1) is the direct target of Bcl3, which showed increased Bcl3 binding to κ B sites in muscles.³⁸ In this study, we found an upregulation of Trim63 expression with OCT treatment (Figures 8A and 11A). Trim63 mutations were described as a potential cause for cardiomyopathy and skeletal myopathy.⁴¹ The inactivation of MuRF1 can lead to an attenuated muscle atrophy response.⁴² Additionally, Trim63 is capable of polyubiquitination and the ubiquitin-proteasome system (UPS)-mediated degradation of thick filament proteins (Myh6 and Mybpc3).⁴³ Mybpc3 belongs to the immunoglobulin Mybp superfamily and contributes to the maintenance of muscle structure, regulation of contraction, and relaxation.⁴⁴ Research studies link truncating mutations in Mybpc3 as the mechanism associated with cardiomyopathy.⁴⁵ The missing Mybpc3 gene leads to a decrease in the myocardial contractility force.⁴⁶ Myh6 is also associated with myocardial contractility.⁴⁷ This study found that OCT decreased the expressions of Mybpc3 and Myh6 (Figures 8B and 11A).

Interestingly, the microarray assay screened out the protein of calcium voltage-gated channel subunit α 1 H (Cacna1h), which could mediate the entry of calcium ions into the

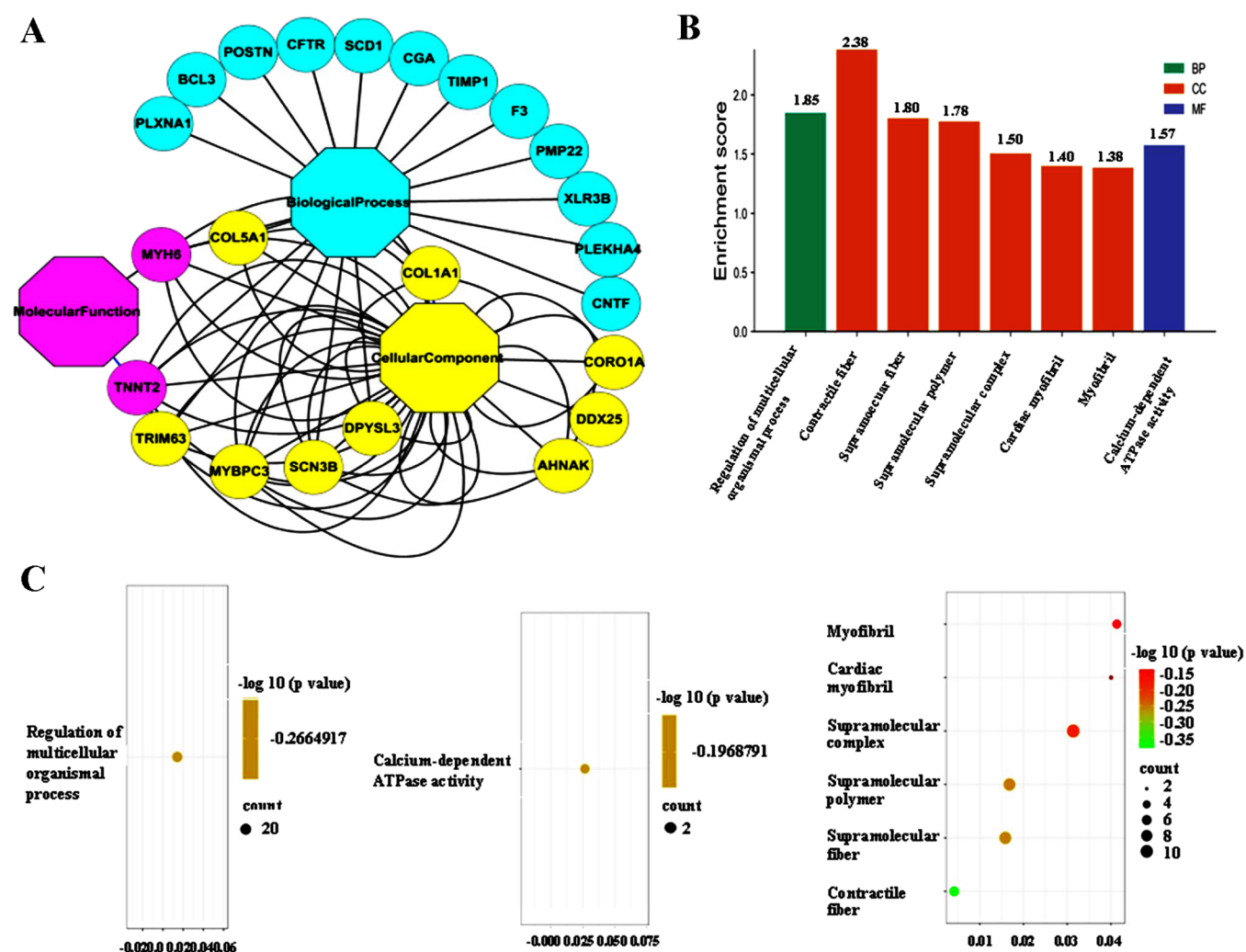


Figure 9. Functional relevance of gene ontology analysis with 67 genes. Interaction network of genes and gene ontology function (A), interrelation analysis between gene ontology terms (B), and bubble chart (C) of gene ontology analysis result. Gene ontology analysis consists of biological processes, cellular components, and molecular functions. The vertical items are the names of gene ontology terms and the length of the horizontal graph represents the adjusted p -value. The depth of the color represents the $-\log_{10}$ adjusted p -value. The area of a circle in the graph means gene counts. These analyses were performed using g:Profiler and Cytoscape 2.8.2 software.

excitable cells and have been implicated in a variety of calcium-dependent processes, including muscle fiber development, muscle contraction, and hormone secretion or neurotransmitter release.^{48–50} Recent publications report that the heterozygous mutation of *Cacna1h* causes congenital muscular atrophy, probably through dysfunctional Ca^{2+} regulation.⁴⁸ Deficiency of *Cacna1h* shows the reduced contractile ability of the muscle.⁴⁹ But our data showed that *Cacna1h* expression was decreased after OCT treatment (Figures 8B and 11A). OCT inhibitory effect is an important factor in protecting myotubes from atrophy and apoptosis induced by T-channel. These findings provided insights into the role of *Cacna1h* as a key regulator of skeletal muscle functions.

Furthermore, our research found that the periaxin (Prx) expression level is decreased after forced exercise, but OCT restored it to normalcy (Figures 8B and 11A). Prx, a Schwann cell-specific protein, during the formation of myelin by Schwann cells wrap the axons, specifically participated in the extension, repair, stabilization, and regeneration of the myelin.⁵¹ The deletion or mutation of the Prx gene induces demyelinating neurological Charcot-Marie-tooth disease sub-

type 4F (CMT4F), which then causes convulsions, dyskinesia, and so on.⁵² Therefore, OCT may also regulate exercise damage through the self-neuroprotective process of the organism, stimulating fatigue recovery functions.

In summary, our present study demonstrated that OCT has antifatigue activity in exercise-induced fatigue models and partly inhibited over-exercise-induced muscle injury in gastrocnemius tissues. This study used gene chip technology for the first time to systematically identify the genes involved in the antifatigue effect of OCT and explored the possible interactions between these genes. The results indicate that OCT modulating expressions of *Bcl3*, *Trim63*, *Mybp3*, *Myh6*, *Cacna1h*, and so on, will ameliorate exercise-induced abnormal reaction. These cytokines synergistically promote muscular atrophy by the interaction between inflammatory cells and organs, resulting in disorders on the balance of protein biosynthesis and degradation, and in muscle loss and exercise functional impairment.^{35,53} These findings have broad significance for understanding the genes and pathways that regulate fatigue function. Additionally, it is necessary to further investigate the gene expression profile of OCT and the

Minimum required interaction score: 0.150

Known interactions:

— from curated databases
— experimentally determined

Predicted Interactions:

— gene neighborhood
— gene fusions
— gene co-occurrence

Others:

— text mining
— co-expression
— protein homology

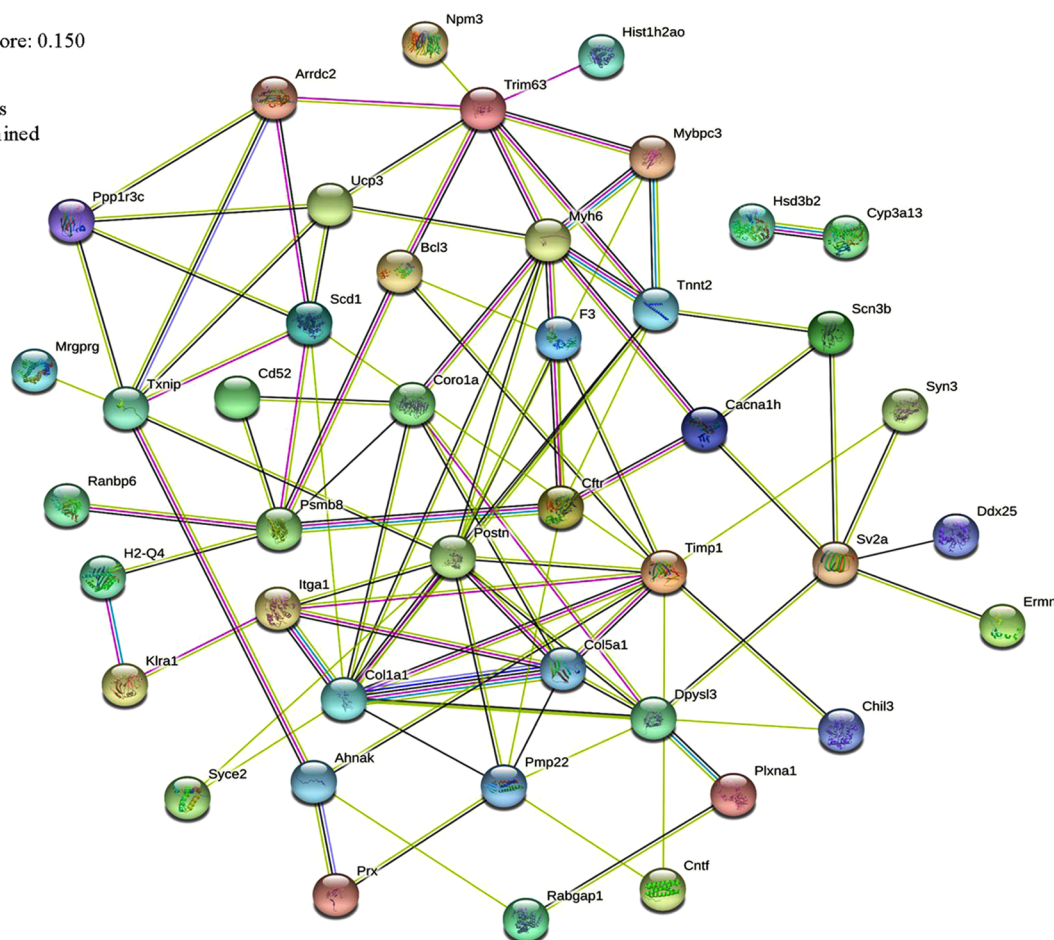


Figure 10. Main network among genes related to fatigue affected by octacosanol. The normalized microarray data were uploaded into STRING database to generate this network through an automatic comparison.

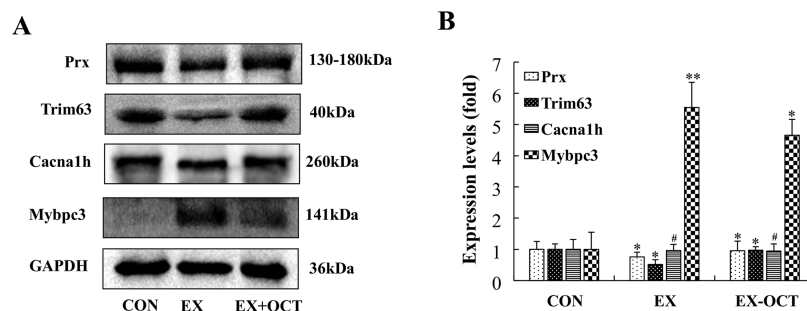


Figure 11. Octacosanol relieves fatigue-related proteins expression levels in forced exercise stimulated in the gastrocnemius tissue of C57BL/6 mice. Mice treated with octacosanol (200 mg/(kg day)) for 30 days. Total protein was extracted from the gastrocnemius tissue of mice after octacosanol treatments, and the expression of Prx, Trim63, Cacna1h, and Mybpc3 proteins was assessed by western blot analysis. The images shown here are representative data from three independent experiments. Values represented as mean $\pm$ SD. ** p < 0.01, * p < 0.05, and # p > 0.05.

interaction between OCT and its target genes. Results from this will advance the understanding of the antifatigue effects of OCT. Taken together, dietary supplements using natural extracts like OCT may be a candidate for adjuvant treatment for fatigue and other related diseases.

PROSPECT

Fatigue is not a disease in medicine and has not strongly attracted the attention of medical researchers. In the high-speed economic society, fatigue almost accompanies our daily life. Since natural products have no side effects or very few side

effects, it is an important idea to screen antifatigue active components from natural products. At present, the research on fatigue and antifatigue in the medical field is still at the initial stage, and the molecular mechanism of fatigue and antifatigue effects of active ingredients is still unclear. In this study, we systematically analyzed the effects of dietary OCT on the gene expression profiling of muscle tissue in the over-exercise mice, identified the important target genes and related signal pathways regulated by OCT, and also identified the network of protein–protein interactions. The study can provide a reference for the study of the antifatigue mechanism of other active components.

This study identified important genes and signaling pathways modulated by OCT, but how does dietary OCT regulate gene expression? Do intestinal microbiota and metabolites play a role in antifatigue action? Are microRNAs, long-coding RNAs, circRNAs, and other noncoding genes involved in the antifatigue effect? At present, these problems are not clear and need further investigations in the future.

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Notes

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ABBREVIATIONS

OCT, octacosanol; CON, control; EX, exercise; LG, liver glycogen; MG, muscle glycogen; BLA, blood lactic acid; LDH, lactate dehydrogenase; SOD, superoxide dismutase; GSH-Px, glutathione peroxidase; HRP, horseradish peroxidase; ELISA, enzyme-linked immunosorbent assay; H&E, hematoxylin and eosin; RNA, ribonucleic acid; cDNA, complementary deoxy-ribonucleic acid; ds-cDNA, double-strand cDNA; EDTA, ethylenediaminetetraacetic acid; RMA, robust multichip average; FC, fold change; RT-qPCR, real-time quantitative polymerase chain reaction; PMSF, phenylmethylsulfonyl fluoride; PVDF, poly(vinylidene difluoride); BSA, bovine serum albumin; TBS-T, Tris-buffered saline with Tween 20; Pmp22, peripheral myelin protein 22; Trim63, tripartite motif containing 63; Ulk3, unc-51-like kinase 3; Arrdc2, arrestin domain containing 2; Prx, periaxin; Bcl3, B cell leukemia/lymphoma 3; Cga, glycoprotein hormones, α subunit; Sv2a, synaptic vesicle glycoprotein 2 a; Cacna1h, calcium channel, voltage-dependent, T type, α 1H subunit; Mybpc3, myosin binding protein C; SEM, standard error mean; SD, standard deviation; GO, gene ontology; NF- κ B, nuclear factor kappa-B; MAPK, mitogen-activated protein kinase; TLRs, Toll-like receptors; TNF- α , tumor necrosis factor- α ; CMT4F, Charcot-Marie-Tooth disease subtype 4F

■ REFERENCES

- (1) Nam, S. Y.; Kim, H. M.; Jeong, H. J. Anti-fatigue effect by active dipeptides of fermented porcine placenta through inhibiting the inflammatory and oxidative reactions. *Biomed. Pharmacother.* **2016**, *84*, 51–59.
- (2) Huang, C. C.; Hsu, M. C.; Huang, W. C.; Yang, H. R.; Hou, C. C. Triterpenoid-rich extract from *Antrodia camphorata* improves physical fatigue and exercise performance in mice. *Evidence-Based Complementary Altern. Med.* **2012**, No. 364741.
- (3) Narkhede, A. N.; Jagtap, S. D.; Nirmal, P. S.; Giramkar, S. A.; Nagarkar, B. E.; Kulkarni, O. P.; Harsulkar, A. M. Anti-fatigue effect of Amarkand on endurance exercise capacity in rats. *BMC Complementary Altern. Med.* **2016**, *16*, No. 23.
- (4) Tan, W.; Yu, K. Q.; Liu, Y. Y.; Ouyang, M. Z.; Yan, M. H.; Luo, R.; Zhao, X. S. Anti-fatigue activity of polysaccharides extract from *Radix Rehmanniae Preparata*. *Int. J. Biol. Macromol.* **2012**, *50*, 59–62.
- (5) You, L.; Ren, J.; Yang, B.; Regenstein, J.; Zhao, M. Antifatigue activities of loach protein hydrolysates with different antioxidant activities. *J. Agric. Food Chem.* **2012**, *60*, 12324–12331.
- (6) Finsterer, J.; Mahjoub, S. Z. Fatigue in healthy and diseased individuals. *Am. J. Hosp. Palliative Care* **2014**, *31*, 562–575.
- (7) Xu, M.; Liang, R.; Li, Y.; Wang, J. Anti-fatigue effects of dietary nucleotides in mice. *Food Nutr. Res.* **2017**, *61*, No. 1334485.
- (8) Chi, A.; Li, H.; Kang, C.; Guo, H.; Wang, Y.; Guo, F.; Tang, L. Anti-fatigue activity of a novel polysaccharide conjugates from Ziyang green tea. *Int. J. Biol. Macromol.* **2015**, *80*, 566–572.
- (9) Taylor, J. C.; Rapport, L.; Lockwood, G. B. Octacosanol in human health. *Nutrition* **2003**, *19*, 192–195.
- (10) Kato, S.; Karino, K.-I.; Hasegawa, S.; et al. Octacosanol affects lipid metabolism in rats fed on a high-fat diet. *Br. J. Nutr.* **1995**, *73*, 433–441.
- (11) Kim, H.; Park, S.; Han, D. S.; Park, T. Octacosanol supplementation increases running endurance time and improves biochemical parameters after exhaustion in trained rats. *J. Med. Food* **2003**, *6*, 345–351.
- (12) Lee, S. H.; Scott, S. D.; Pekas, E. J.; Lee, J. K.; Park, S. Y. Improvement of lipids and reduction of oxidative stress with octacosanol after taekwondo training. *Int. J. Sports Physiol.* **2019**, 1297–1303.
- (13) Menchetti, L.; Guelfi, G.; Speranza, R.; Carotenuto, P.; Moscatti, L.; Diverio, S. Benefits of dietary supplements on the physical fitness of German Shepherd dogs during a drug detection training course. *PLoS One* **2019**, *14*, No. e0218275.
- (14) Wang, T.; Liu, Y.; Yang, N.; Ji, C.; Chan, P.; Zuo, P. Anti-parkinsonian effects of octacosanol in 1-methyl-4-phenyl-1, 2, 3, 6 tetrahydropyridine-treated mice. *Neural Regen. Res.* **2012**, *7*, 1080–1087.
- (15) Thippeswamy, G.; Sheela, M. L.; Salimath, B. P. Octacosanol isolated from *tinospora cordifolia* downregulates VEGF gene expression by inhibiting nuclear translocation of NF-kappa B and its DNA binding activity. *Eur. J. Pharmacol.* **2008**, *588*, 141–150.
- (16) de Oliveira, A. M.; Conserva, L. M.; De souza ferro, J. N.; De Almeida Brito, F.; Lyra Lemos, R. P.; Barreto, E. Antinociceptive and anti-inflammatory effects of octacosanol from the leaves of *Sabicea grisea* var. *grisea* in mice. *Int. J. Mol. Sci.* **2012**, *13*, 1598–1611.
- (17) Kaushik, M. K.; Aritake, K.; Takeuchi, A.; Yanagisawa, M.; Urade, Y. Octacosanol restores stress-affected sleep in mice by alleviating stress. *Sci. Rep.* **2017**, *7*, No. 8892.
- (18) Sharma, R.; Matsuzaka, T.; Kaushik, M. K.; Sugawara, T.; Ohno, H.; Wang, Y.; Motomura, K.; Shimura, T.; Okajima, Y.; Mizunoe, Y.; Ma, Y.; Saber, Z. M.; Iwasaki, H.; Yatoh, S.; Suzuki, H.; Aita, Y.; Han, S. I.; Takeuchi, Y.; Yahagi, N.; Miyamoto, T.; Sekiya, M.; Nakagawa, Y.; Shimano, H. Octacosanol and policosanol prevent high-fat diet-induced obesity and metabolic disorders by activating brown adipose tissue and improving liver metabolism. *Sci. Rep.* **2019**, *9*, No. 5169.
- (19) Jiang, M. Y.; Lu, H.; Pu, X. Y.; Li, Y. H.; Tian, K.; Xiong, Y.; Wang, W.; Huang, X. Z. Laxative metabolites from the leaves of *moringa oleifera*. *J. Agric. Food Chem.* **2020**, *68*, 7850–7860.
- (20) Guo, T.; Lin, Q.; Li, X.; Nie, Y.; Wang, L.; Shi, L.; Xu, W.; Hu, T.; Guo, T.; Luo, F. Octacosanol attenuates inflammation in both RAW264.7 macrophages and a mouse model of colitis. *J. Agric. Food Chem.* **2017**, *65*, 3647–3658.
- (21) Jiang, X.; Chu, Q.; Li, L.; Qin, L.; Hao, J.; Koum, L.; Lin, F.; Wang, D. The anti-fatigue activities of tuber *melanosporum* in a mouse model. *Exp. Ther. Med.* **2018**, *15*, 3066–3073.
- (22) Park, S. H.; Jang, S.; Lee, S. W.; Park, S. D.; Sung, Y. Y.; Kim, H. K. *Akebia quinata* Decaisne aqueous extract acts as a novel anti-fatigue agent in mice exposed to chronic restraint stress. *J. Ethnopharmacol.* **2018**, *222*, 270–279.
- (23) Wei, Y. F.; Tao, H. H. Effect of polysaccharides from the roots of *Morinda officinalis* how on physical fatigue. *J. Food Agric. Environ.* **2013**, *11*, 581–584.
- (24) Huang, W. C.; Chiu, W. C.; Chuang, H. L.; Tang, D. W.; Lee, Z. M.; Wei, L.; Chen, F. A.; Huang, C. C. Effect of curcumin supplementation on physiological fatigue and physical performance in mice. *Nutrients* **2015**, *7*, 905–921.
- (25) Ding, J. F.; Li, Y. Y.; Xu, J. J.; Su, X. R.; Gao, X.; Yue, F. P. Study on effect of jellyfish collagen hydrolysate on anti-fatigue and anti-oxidation. *Food Hydrocolloids* **2011**, *25*, 1350–1353.
- (26) Gibson, H.; Edwards, R. H. Muscular exercise and fatigue. *Sports Med.* **1985**, *2*, 120–132.
- (27) Song, H.; Jang, M. G.; Lee, J. Y.; Ko, H. C.; Hur, S. P.; Kim, S. J. Sasa quelpaertensis leaf extract mitigates fatigue and regulates the transcriptome profile in mice. *Genes Genomics* **2019**, *41*, 317–324.
- (28) Kabir, Y.; Kimura, S. Distribution of radioactive octacosanol in response to exercise in rats. *Nahrung* **1994**, *38*, 373–377.
- (29) Zhao, R.; Cai, Y.; Shao, X.; Ma, B. Improving the activity of *Lycium barbarum* polysaccharide on sub-health mice. *Food Funct.* **2015**, *6*, 2033–2040.
- (30) Yan, F.; Zhang, Y.; Wang, B. B. Effects of polysaccharides from *Cordyceps sinensis* mycelium on physical fatigue in mice. *Bangladesh J. Pharmacol.* **2012**, *7*, 217–221.
- (31) Zhang, G.; Zhou, S. M.; Tian, J. H.; Huang, Q. Y.; Gao, Y. Q. Anti-fatigue effects of methazolamide in high-altitude hypoxic mice. *Trop. J. Pharm. Res.* **2012**, *11*, 209–215.
- (32) Gladden, L. B. Lactate metabolism: a new paradigm for the third millennium. *J. Physiol.* **2004**, *558*, 5–30.
- (33) Yu, S. H.; Huang, H. Y.; Korivi, M.; Hsu, M. F.; Huang, C. Y.; Hou, C. W.; Chen, C. Y.; Kao, C. L.; Lee, R. P.; Lee, S. D.; Kuo, C. H. Oral Rg1 supplementation strengthens antioxidant defense system against exercise-induced oxidative stress in rat skeletal muscles. *J. Int. Soc. Sports Nutr.* **2012**, *9*, No. 23.
- (34) Herrington, F. D.; Nibbs, R. J. Regulation of the adaptive immune response by the IκB family protein Bcl-3. *Cells* **2016**, *5*, No. 14.
- (35) Baumert, P.; G-REX Consortium; Lake, M. J.; Drust, B.; Stewart, C. E.; Erskine, R. M. TRIM63 (MuRF-1) gene polymorphism is associated with biomarkers of exercise-induced muscle damage. *Physiol. Genomics* **2018**, *50*, 142–143.
- (36) Jackman, R. W.; Wu, C. L.; Kandarian, S. C. The ChIP-seq-defined networks of Bcl-3 gene binding support its required role in skeletal muscle atrophy. *PLoS One* **2012**, *7*, No. e51478.
- (37) Hunter, R. B.; Kandarian, S. C. Disruption of either the Nfkb1 or the Bcl3 gene inhibits skeletal muscle atrophy. *J. Clin. Invest.* **2004**, *114*, 1504–1511.
- (38) Wu, C. L.; Kandarian, S. C.; Jackman, R. W. Identification of genes that elicit disuse muscle atrophy via the transcription factors p50 and Bcl-3. *PLoS One* **2011**, *6*, No. e16171.
- (39) Collins, P. E.; Somma, D.; Kerrigan, D.; Herrington, F.; Keeshan, K.; Nibbs, R. J. B.; Carmody, R. J. The IκB-protein BCL-3 controls Toll-like receptor-induced MAPK activity by promoting TPL-2 degradation in the nucleus. *Proc. Natl. Acad. Sci. U.S.A.* **2019**, *116*, 25828–25838.
- (40) Li, Y. P.; Reid, M. B. NF-kappaB mediates the protein loss induced by TNF-alpha in differentiated skeletal muscle myotubes. *Am. J. Physiol.: Regul., Integr. Comp. Physiol.* **2000**, *279*, R1165–R1170.

- (41) Jokela, M.; Baumann, P.; Huovinen, S.; Penttilä, S.; Udd, B. Homozygous nonsense mutation p.Q274x in Trim63 (MuRF1) in a patient with mild skeletal myopathy and cardiac hypertrophy. *J. Neuromuscular Dis.* **2019**, *6*, 143–146.
- (42) Labeit, S.; Kohl, C. H.; Witt, C. C.; Labeit, D.; Jung, J.; Granzier, H. Modulation of muscle atrophy, fatigue and MLC phosphorylation by MuRF1 as indicated by hindlimb suspension studies on MuRF1-KO mice. *J. Biomed. Biotechnol.* **2010**, No. 693741.
- (43) Clarke, B. A.; Drujan, D.; Willis, M. S.; Murphy, L. O.; Corpina, R. A.; Burova, E.; Rakhilin, S. V.; Stitt, T. N.; Patterson, C.; Latres, E.; Glass, D. J. The E3 ligase MuRF1 degrades myosin heavy chain protein in dexamethasone-treated skeletal muscle. *Cell Metab.* **2007**, *6*, 376–385.
- (44) Carrier, L.; Mearini, G.; Stathopoulou, K.; Cuello, F. Cardiac myosin-binding protein C (MYBPC3) in cardiac pathophysiology. *Gene* **2015**, *573*, 188–197.
- (45) Toepfer, C. N.; Wakimoto, H.; Garfinkel, A. C.; McDonough, B.; Liao, D.; Jiang, J.; Tai, A. C.; Gorham, J. M.; Lunde, I. G.; Lun, M.; Lynch, T. L.; McNamara, J. W.; Sadayappan, S.; Redwood, C. S.; Watkins, H. C.; Seidman, J. G.; Seidman, C. E. Hypertrophic cardiomyopathy mutations in MYBPC3 dysregulate myosin. *Sci. Transl. Med.* **2019**, *11*, No. eaat1199.
- (46) Jiang, J.; Burgon, P. G.; Wakimoto, H.; Onoue, K.; Gorham, J. M.; O'Meara, C. C.; Fomovsky, G.; McConnell, B. K.; Lee, R. T.; Seidman, J. G.; Seidman, C. E. Cardiac myosin binding protein C regulates postnatal myocyte cytokinesis. *Proc. Natl. Acad. Sci. U.S.A.* **2015**, *112*, 9046–9051.
- (47) Tomita-Mitchell, A.; Stamm, K. D.; Mahnke, D. K.; Kim, M. S.; Hidestrand, P. M.; Liang, H. L.; Goetsch, M. A.; Hidestrand, M.; Simpson, P.; Pelech, A. N.; Tweddell, J. S.; Benson, D. W.; Lough, J. W.; Mitchell, M. E. Impact of MYH6 variants in hypoplastic left heart syndrome. *Physiol. Genomics* **2016**, *48*, 912–921.
- (48) Carter, M. T.; Mcmillan, H. J.; Tomin, A.; Weiss, N. Compound heterozygous CACNA1H mutations associated with severe congenital amyotrophy. *Channels* **2019**, *13*, 153–161.
- (49) Li, S.; Hao, M.; Li, B.; Chen, M.; Chen, J.; Tang, J.; Hong, S.; Min, J.; Hu, M.; Hong, L. CACNA1H downregulation induces skeletal muscle atrophy involving endoplasmic reticulum stress activation and autophagy flux blockade. *Cell Death Dis.* **2020**, *11*, No. 279.
- (50) Stringer, R. N.; Jurkovicova-Tarabova, B.; Huang, S.; Haji-Ghassemi, O.; Idoux, R.; Liashenko, A.; Souza, I. A.; Rzhetsky, Y.; Lacinova, L.; Van Petegem, F.; Zamponi, G. W.; Pamphlett, R.; Weiss, N. A rare CACNA1H variant associated with amyotrophic lateral sclerosis causes complete loss of Ca_v3.2 T-type channel activity. *Mol. Brain* **2020**, *13*, No. 33.
- (51) Raasakka, A.; Linxweiler, H.; Brophy, P. J.; Sherman, D. L.; Kursula, P. Direct binding of the flexible C-terminal segment of periaxin to β 4 integrin suggests a molecular basis for CMT4F. *Front. Mol. Neurosci.* **2019**, *12*, No. 84.
- (52) Sherman, D. L.; Brophy, P. J. A murine model of Charcot-Marie-Tooth disease 4F reveals a role for the C-terminus of periaxin in the formation and stabilization of Cajal bands. *Wellcome Open Res.* **2018**, *3*, No. 20.
- (53) Majumder, A.; Singh, M.; Behera, J.; Theilen, N. T.; George, A. K.; Tyagi, N.; Metreveli, N.; Tyagi, S. C. Hydrogen sulfide alleviates hyperhomocysteinemia-mediated skeletal muscle atrophy via mitigation of oxidative and endoplasmic reticulum stress injury. *Am. J. Physiol.: Cell Physiol.* **2018**, *315*, C609–C622.